higher education & training

Department:
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REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE

INDUSTRIAL ELECTRONICS N5

(8080175)

4 April 2024 (X-paper)
09:00–12:00

Drawing instruments and nonprogrammable calculators may be used.

This question paper consists of 6 pages and a formula sheet of 6 pages.

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DEPARTMENT OF HIGHER EDUCATION AND TRAINING
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE
INDUSTRIAL ELECTRONICS N5
TIME: 3 HOURS
MARKS: 100

INSTRUCTIONS AND INFORMATION

1. Answer all the questions.
2. Read all the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Use only a black or blue pen.
5. Write neatly and legibly.

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QUESTION 1: ALTERNATING CURRENT THEORY

1.1 With the aid of a neat, labelled R-C integrator diagram, indicate this type of passive filter. (3)

1.2 Choose the correct word or words from those given in brackets. Write only the word or words next to the question number (1.2.1–1.2.2) in the ANSWER BOOK.

1.2.1 In an RC-integrated circuit the mark period of the applied square waveform is (greater/smaller) than the RC-time constant.

1.2.2 (Square waveforms/Sine wave forms) indicate the presence as well as the type of error during the testing of electronic circuits.

(2 × 1) (2)

1.3 The voltage phasor diagram of a series RLC-circuit has the following values:

$$V_L = 242{,}5 \angle 38{,}32^0 \text{ V}$$

$$V_C = 164 \angle -141{,}68^0 \text{ V}$$

$$V_R = 61{,}8 \angle -51{,}68^0 \text{ V}$$

$$I_T = 5{,}15 \angle -51{,}68^0 \text{ A}$$

$$V_T = 100 \angle 0^0 \text{ V}$$

1.3.1 Draw the phasor diagram. (3)

1.3.2 Calculate:

A The reactance of the coil
B The reactance of the capacitor
C The resistance of the series resistor

(3 × 2) (6)
[14]

QUESTION 2: POWER SUPPLIES

2.1 The power supply of a Samsung TV uses a π-type filter with an input DC voltage of 18 V across the first capacitor C_1. The filter inductor has an inductance of 5 H and a wire resistance of 20 Ω.

Calculate the output DC voltage at a load current of 50 mA. (2)

2.2 Explain the reason why the capacitors in the filter circuit of a power supply are connected in parallel and not in series with the load. (2)

2.3 Calculate the drop in the output voltage if a power supply of 25 V (no-load voltage) has a voltage regulation of 8% as soon as it is connected to the load. (3)

2.4 Draw a neat, labelled circuit diagram of a shunt voltage regulator and briefly describe its operation. (7)

[14]

QUESTION 3: TRANSISTOR AMPLIFIERS

3.1 Briefly describe the functions of the following components in a common emitter amplifier used on a communication system:

3.1.1 The voltage divider R_{B1} and R_{B2}

3.1.2 The input capacitor C_B

(2 × 2) (4)

3.2 The following values of a fixed-bias voltage amplifier (without thermal components) are known:

$^{+}V_{CC}$ = 20 V; I_C = 14 mA; R_C = 1 kΩ; R_B = 800 kΩ; V_{BE} =0,7 V

Calculate the following values:

3.2.1 The collector-emitter voltage (3)

3.2.2 The base current (3)

3.2.3 The beta gain (2)

3.3 A supersonic radio has a fixed-bias voltage with an input impedance of 1,169 k Ω and an output impedance of 1,96 k Ω.

Calculate the values of the collector resistor and base resistor with the precise method if the following information of the transistor is known:

h_{ie}=1,2 kΩ; h_{oe}=20 µA/V; $h_{re} = 2 \times 10^{-4}$; $hf_e = 60$ $R_S = 0$ (6)

[18]

QUESTION 4: OPERATIONAL AMPLIFIERS

4.1 Briefly explain the following terms with reference to operational amplifiers:

4.1.1 Input offset voltage

4.1.2 Inverting input

(2 × 2) (4)

4.2 4.2.1 Draw the circuit diagram of an inverting operational amplifier. (3)

4.2.2 The input signal to an operational amplifier in the inverting mode is -2 V and the input resistance is 15 kΩ. If the output voltage is 20 V, calculate the feedback resistor. (3)

[10]

QUESTION 5: INTEGRATED CIRCUITS

5.1 What should be done with the unused inputs of a CMOS integrated circuit when it is connected to a circuit? (2)

5.2 Indicate whether the following statements are TRUE or FALSE by writing only 'True' or 'False' next to the question number (5.2.1–5.2.3) in the ANSWER BOOK.

5.2.1 CMOS-integrated circuits have lower noise immunity.

5.2.2 CMOS-integrated circuits are susceptible to static charges because of their low reactive input.

5.2.3 As soon as one works on the circuits of CMOS-integrated circuits, the power supply to the circuit must be switched on.

(3 × 1) (3)

[5]

QUESTION 6: ELECTRONIC PHASE CONTROL

Draw a neatly labelled block diagram of a general closed-loop system and briefly describe how the load condition is kept constant. **[8]**

QUESTION 7: TEST EQUIPMENT

With the aid of a neat, labelled block diagram, draw an instrument that displays the frequency of periodic electrical signal. **[5]**

QUESTION 8: OSCILLATORS

8.1 If the total inductance of a Hartley tuned circuit is 10 mH and the oscillating frequency is 300 kHz, what is the value of the capacitor in the tuned circuit? (3)

8.2 Draw a neat, labelled circuit diagram of a 555-timer in an astable operation. (4)

8.3 Calculate the oscillation frequency of an astable 555-timer with the following values:

R_A = 90 kΩ

R_B =60 kΩ

C = 1 μf

(3 × 1) (3)

8.4 Name TWO factors that may destabilise the frequency of an oscillator. (2)

[12]

QUESTION 9: TRANSDUCERS

9.1 9.1.1 Draw a neat, labelled circuit diagram of a closed-loop system in which synchronous motors are used to control a radar antenna. (5)

9.1.2 Briefly describe what happens as soon as the transmitter rotor moves through an angle θ. (4)

9.2 Draw a neat, labelled circuit diagram of a precision switch in which a light dependant resistor and an operational amplifier are used. (5)

[14]

100

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FORMULA SHEET

$$I = \frac{V}{R} \qquad P = IV = I^2R = \frac{V^2}{R}$$

$$V_T = V_1 + V_2 + V_3 + ... = I_1R_1 + I_2R_2 + I_3R_3 + ...$$

$$I_T = I_1 + I_2 + I_3 + ... = \frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + ...$$

$$T = RC \qquad T = \frac{L}{R}$$

$$V_R = RC\frac{dv}{dt} \qquad V_C = \frac{1}{RC}\int v_i dt$$

$$X_L = 2\pi fL \qquad X_C = \frac{1}{2\pi fC}$$

$$Z = R + jX_L \qquad Z = R - jX_C$$

$$Z = R + j(X_L - X_C) \qquad I_T = \frac{V_T}{Z_T}$$

$$V_R = I_TR \qquad V_L = I_T(jX_L)$$

$$V_C = I_T(-jX_C) \qquad f_r = \frac{1}{2\pi\sqrt{LC}}$$

$$Q = \frac{V_L}{V_T} = \frac{V_C}{V_T} = \frac{X_L}{R} = \frac{X_C}{R} = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{f_r}{f_2 - f_1}$$

$$BW = f_2 - f_1 \qquad \frac{1}{Z_T} = \frac{1}{Z_1} + \frac{1}{Z_2}$$

$$Z_T = \frac{Z_2Z_2}{Z_1 + Z_2} \qquad I_T = I_1 + I_2 = \frac{V}{Z_1} + \frac{V}{Z_2}$$

$$Z_T = \frac{R(jX_L)}{R + jX_L} \qquad \frac{1}{Z_T} = \frac{1}{R} - \frac{j}{X_L}$$

$$I_T = I_R - jI_L \qquad I_T = \frac{V}{R} - j\frac{V}{X_L}$$

$$Z_T = \frac{R(-jX_C)}{R - jX_C} \qquad \frac{1}{Z_T} = \frac{1}{R} + \frac{j}{X_C}$$

$$I_T = I_R + jI_C \qquad I_T = \frac{V}{R} + j\frac{V}{X_C}$$

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$$\frac{1}{Z_T} = \frac{1}{R} - j\left(\frac{1}{X_L} - \frac{1}{X_C}\right)$$

$$I_T = I_R - j(I_L - I_C)$$

$$I_T = \frac{V}{R} - j\left(\frac{V}{X_L} - \frac{V}{X_C}\right)$$

$$a + jb = \sqrt{a^2 + b^2} \,/\, \tan^{-1}\frac{b}{a} = r \,\underline{/\theta}$$

$$r\,\underline{/\theta} = r(\cos\theta + j\sin\theta)$$

$$Q = \tan\theta$$

$$f = \frac{1}{2\pi}\sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$Z_d = \frac{L}{CR_1}$$

$$V_{rms} = \frac{1}{\sqrt{2}}V_m = 0{,}707\,V_m$$

$$V_{dc} = \frac{2}{\pi}V_m = 0{,}637\,V_m$$

$$\frac{V_P}{V_S} = \frac{N_P}{N_S} = \frac{I_S}{I_P}$$

$$V_{dc} = \frac{1}{\pi}V_m = 0{,}318\,V_m$$

$$PIV = V_m$$

$$PIV = 2\,V_m$$

$$R_{r(rms)} = 0{,}385\,V_m$$

$$V_{r(rms)} = 0{,}305\,V_m$$

$$r = \frac{V_{r(rms)}}{V_{dc}}$$

$$V_{r(rms)} = \frac{V_{r(p-p)}}{2\sqrt{3}}$$

$$V_{dc} = V_m - \frac{V_{r(p-p)}}{2}$$

$$V_{dc} = V_m - \frac{I_{dc}}{2fC}$$

$$V_{dc} = V_m - \frac{I_{dc}}{4fC}$$

$$V_{r(rms)} = \frac{I_{dc}}{2\sqrt{3}fC} = \frac{V_{dc}}{2\sqrt{3}fCR_L}$$

$$V_{r(rms)} = \frac{I_{dc}}{4\sqrt{3}fC} = \frac{V_{dc}}{4\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{2\sqrt{3}fCV_{dc}} = \frac{1}{2\sqrt{3}fCR_L}$$

$$r = \frac{I_{dc}}{4\sqrt{3}fCV_{dc}} = \frac{1}{4\sqrt{3}fCR_L}$$

$$V'_{dc} = \frac{R_L}{R + R_L}.V_{dc}$$

$$V'_{r(rms)} = \frac{X_C}{\sqrt{R^2 + X_C^2}}\cdot V_{r(rms)}$$

$$X_C = \frac{1}{2\pi fC} \qquad X_C = \frac{1}{4\pi fC} \qquad r' = \frac{V'_{r(rms)}}{V'_{dc}}$$

$$V'_{r(rms)} = \frac{X_C}{R}.V_{r(rms)} \qquad r' = rX_C\left(\frac{R + R_L}{R.R_L}\right)$$

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$$TV'_{dc} = V_{dc} - I_{dc}R_1$$

$$V'_{dc} = \frac{R_L}{R_L + R_1} . V_{dc}$$

$$V'_{r(rms)} = \frac{V_{r(rms)}}{(2\pi f)^2 LC}$$

$$V'_{r(rms)} = \frac{V_{r(rm)}}{(4\pi f)^2 LC}$$

$$VR = \frac{V_{NL} - V_{FL}}{V_{FL}}$$

$$\%VR = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100$$

$$2V_m = V_{c2} = V_m + V_{c1}$$

$$3V_m = V_{c1} + V_{c3} = V_m + 2V_m$$

$$S = \frac{\Delta V_o}{\Delta V_i}$$

$$V_R = V_i - V_z$$

$$R_{s(\min)} = \frac{V_{i(\max)} - V_z}{I_{z(\max)}}$$

$$I_z = \frac{P_z}{V_z}$$

$$R_{L(\min)} = \frac{V_Z}{V_{i(\max)} - V_Z} . R_S$$

$$V_o = V_r - V_{be}$$

$$R_c = \frac{V_{cc} - V_{ce}}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be}}{I_b}$$

$$\beta = \frac{I_c}{I_b}$$

$$C_e \geq \frac{10}{2\pi f R_e}$$

$$V_e = \frac{V_{cc}}{10}$$

$$R_e = \frac{V_e}{I_e} \simeq \frac{V_e}{I_c}$$

$$R_c = \frac{V_{cc} - V_{ce} - V_e}{I_c}$$

$$R_b = \frac{V_{cc} - V_{be} - V_e}{I_b}$$

$$R_{b1} = \frac{R_{b2}(V_{cc} - V_b)}{V_b}$$

$$R_{b2} = \frac{1}{10}\beta R_e$$

$$V_b = V_e + V_{be}$$

$$V_{be} = h_{ie} i_b + h_{re} V_{ce}$$

$$i_c = i_{fe} i_b + h_{oe} V_{ce}$$

$$A_i = \frac{h_{fe}}{1 + h_{oe} Z_L}$$

$$A_i = h_{fe}$$

$$A_i = \left(\frac{h_{fe}}{1 + h_{oe} Z_L}\right)\left(\frac{R_b T}{R_{bT} + Z_1}\right)\left(\frac{R_c}{R_c + R_L}\right)$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie} + (h_{ie} h_{oe} - h_{fe} h_{re}) Z_L}$$

$$A_v = \frac{-h_{fe} Z_L}{h_{ie}}$$

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$$Z_1 = h_{ie} - \frac{h_{fe} h_{re} Z_L}{1 + h_{oe} Z_L} \qquad Z_1 = h_{ie}$$

$$Z_2 = \frac{1}{h_{oe} - \frac{h_{fe} h_{re}}{h_{ie} + R_s}} \qquad Z_2 = \frac{1}{h_{oe}}$$

$$A_p = \frac{A_i^2 R_L}{R_1} = -A_v A_i \qquad A_p = \frac{{h_{fe}}^2 R_L}{h_{ie}}$$

$$Z_0 = R_C // R_L // Z_2 = Z_L // Z_2$$

$$Z_0 = R_C // Z_2 = Z_L // Z_2 \qquad Z_1 = R_b // Z_1$$

$$Z_i = R_{b1} // R_{b2} // Z_1 \qquad I_1 = \frac{R_{bT} I_i}{R_{bT} = Z_1}$$

$$I_0 = h_{fe} I_b = h_{fe} \left(\frac{R_{b2}(I_i)}{R_{b2} + h_{ie}} \right) \qquad A_i = \frac{I_0}{I_1}$$

For common base, substitute all the 'e' subscripts with a 'b' in the h-parameters.

$$Z_L = R_c // R_L \qquad I_1 = \frac{R_e I_i}{R_e + Z_1}$$

$$CMRR = \frac{A_{dm}}{A_{cm}} \qquad CMRR\ (dB) = 20log \frac{A_{dm}}{A_{cm}}$$

$$I_e = \frac{V_e}{R_e} \qquad I_c = \frac{I_e}{2}$$

$$R_L = \frac{V_{R_L}}{I_C} \qquad g_m R_L = \frac{h_{fe}}{h_{ie}} . R_L$$

$$V_0 = -\left(\frac{R_f}{R_1} \right) . V_i \qquad V_0 = \left(\frac{R_f}{R_1} + 1 \right) . V_i$$

$$V_0 = -\left(\frac{R_f}{R_1} . V_1 + \frac{R_f}{R_2} . V_2 + \frac{R_f}{R_3} . V_3 \right) \quad V_0 = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} \right) R_f$$

$$V_0 = -(V_1 + V_2 + V_3) \qquad V_0 = -(I_1 + I_2 + I_3)\ R_f$$

$$V_0(t) = -\frac{1}{RC} \int V_i(t) \qquad V_0\ (t_b) = -\frac{1}{RC} \int_{t_a}^{t_b} V_i(t_b) + V_c(t_a)$$

$$t = \frac{1}{f} \qquad A_v = -\frac{R_s}{R_1}$$

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$$R_2 = \frac{R_1 R_s}{R_1 + R_s}$$

$$f_c = \frac{1}{2\pi R_s C}$$

$$V_0(t) = -RC\frac{dV_i(t)}{dt}$$

$$A = \frac{R_f}{X_c}$$

$$A = -\frac{R_f}{R_s}$$

$$V_0(t) = -R_f C\frac{d}{dt}.v_i \sin \omega t$$

$$t = R_f C$$

$$V_0 = A(V_r - V_i)$$

$$V_0 = \frac{R_f}{R_s}(V_2 - V_1)$$

$$V_0 = V_2 - V_1$$

$$f_0 = \frac{1}{2\pi\sqrt{C_1 C_2 R_1 R_2}}$$

$$f_0 = \frac{1}{2\pi RC}$$

$$f_0 = \frac{1}{2\pi\sqrt{L_T C_1}}$$

$$L_T = L_1 + L_2 + 2M$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_T}}$$

$$C_T = \frac{C_1 C_2}{C_1 + C_2}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC_2}}$$

$$f = \frac{1}{2\pi RC\sqrt{6}}$$

$$f_0 = \frac{1,5}{RC}$$

$$f_0 = \frac{1}{t} = \frac{1}{t_1 + t_2}$$

$$t_1 = 0,7\ R_2 C_1$$

$$t_2 = 0,7\ R_1 C_2$$

$$f_0 = \frac{1}{1,4RC}$$

$$V_i = I_{c2}R_e + V_{be(ON)}$$

$$t = 1,1\ RC$$

$$f_0 = \frac{1,443}{(R_A + 2R_B)C}$$

$$t_{low} = 0,693\,(R_B)C$$

$$t_{high} = 0,693\,(R_A + R_B)C$$

$$t_T = t_{low} + t_{high}$$

$$K = \frac{\Delta R / R}{\Delta l / l}$$

$$\sigma = \Delta l / l$$

$$R = \rho\frac{1}{\pi d^2 / 4}$$

$$\sigma = \frac{S}{E}$$

$$Resolution = \frac{1}{amount\ of\ turns}$$

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$$Resolution = \frac{voltage\ drop\ across\ adjacent\ turns}{total\ voltage\ drop}$$

$R_t = Ae^{B/T}$ $T = 273 + °C$

$V_A = \frac{R_2}{R_1 + R_2} . V_T$ $V_B = \frac{R_t}{R_t + R_3} . V_T$

$V_{AB} = V_A - V_B$ $A_v = \frac{V_0}{V_i}$

$V_{Hall} = kIH$

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